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Activity propagation with Hebbian learning



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Biological and social systems, including infection spreading, inter-regional brain activity propagation, and population spreading, exhibit learning across a broad range of scales. These applications of the contact process therefore call for an extension that incorporates local learning rules. Here we introduce learning as a positive (Hebbian) or negative (anti-Hebbian) reinforcement of the activation rate between a pair of sites after each successful activation event. We show that Hebbian learning leads to a rich class of emergent behaviors, where local incentives can produce opposite global effects. In general, positive reinforcement causes the loss of the active phase, while negative reinforcement can turn the inactive phase into a globally active phase. Our analytical and numerical results demonstrate that, in two dimensions and above, the effect of negative reinforcement is twofold: it promotes the spreading of activity while simultaneously generating effectively immune regions, leading to the emergence of two distinct critical points. By contrast, positive reinforcement can give rise to Griffiths effects with non-universal power-law scaling, a manifestation of the ‘ant-mill’ phenomenon.

Biological and social systems show learning and adaptation across a broad range of scales^{1–3}. Yet, understanding the impact of local learning mechanisms remains mostly unexplored even in simple models⁴. Here, we focus on a paradigmatic model of activity (or infection) spreading, the contact process (CP)⁵ and explore the impact of local learning on its dynamics, motivated by Hebbian learning⁶. As introduced by T. E. Harris, the CP is a variant of the susceptible-infected-susceptible (SIS) model that is equivalent to the standard SIS model on regular lattices⁵. The CP is a continuous-time Markov process that acts on a network of sites with each site either in an “active” or “inactive” state. Other common terminology for these states are infected/healthy and occupied/unoccupied. Each site can be active multiple times, like how a person can get sick many times in the case of the common cold or COVID, or how a neuron or brain region can be active on multiple occasions.

In the case of infection spreading, it is natural to explore what happens if individuals learn from the dynamics and lower the rate at which they interact with those individuals that infected them, a form of negative reinforcement, or anti-Hebbian learning⁷. For example, a reinforcement effect similar to social distancing⁸ would result in a mutual learning mechanism, where the two sites involved would reduce the time spent together.

In the context of brain dynamics, each site can be viewed as a small region of the brain (on the order of a mm³) that can be either active or

inactive and each edge can be viewed as a connection between the brain regions⁹. As chemical synapses in the brain are directed, this motivates implementing a directed learning mechanism, either impacting the connection starting at the source or at the target. Positive reinforcement between brain regions aligns with the observation that brain regions that fire together will wire together more strongly^{10–12}.

As a further motivation, the CP has been also applied to the spreading of populations¹³. For example, parasites spreading on a background of hosts can induce local adaptations through a triggered immune response and resistance resulting from the host-parasite “arms race”^{14–17}, leading to a form of negative reinforcement. Positive reinforcement could also happen along ecological corridors that are strips of favorable conditions (e.g., land and water) connecting fragmented habitats¹⁸. Once established, these corridors can be preferably used later by the population.

Both negative and positive reinforcement leads to heterogeneities in the system if started from a homogeneous network, a form of kinetic disorder. Here, we address the question of how local learning changes the large-scale emergent behavior of the system, compared to the standard CP. The standard CP is only a rough approximation of the aforementioned biological systems, but it is extremely well understood as a starting point. Here, we seek to study the fundamental impact of local learning rules on this paradigmatic process, focusing on the emergent mechanisms, instead of creating a realistic

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model for a particular system. To begin, we provide a brief recap of the standard CP. The standard CP has two phases, an active phase and an inactive phase, governed by the activation rate λ . In the active phase, the infection survives in an infinite system, whereas, in the inactive phase, the infection dies out and never returns—the system reaches an absorbing state. This is known as an absorbing phase transition^{19,20}. The transition in the standard CP serves as a representative of the robust directed percolation universality class (DP)^{21,22}, which is believed to contain any absorbing phase transition with short-range interactions, a scalar order parameter, a single absorbing state, and no extra conservation laws or symmetries^{23,24}. Outside of the CP, transitions of the DP class occur in a broad range of systems, including models of catalytic reactions²⁵, damage spreading transitions²⁶ and interface growth²⁷.

At the critical activation rate of the transition, labeled λ_c , the standard CP follows a power-law behavior for long times. When starting from a single active site on an infinite d -dimensional lattice, the number of active sites N , survival probability P , and the root-mean-square distance of the active sites from the initial active site R scale asymptotically as^{19,20}

$$N \sim t^\eta \quad (1)$$

$$P \sim t^{-\delta} \quad (2)$$

$$R \sim t^{1/z} \quad (3)$$

with critical exponents η , δ , and $1/z$. The standard CP also showcases a phenomenon called duality^{28,29}. When starting from a fully-active lattice with periodic boundary conditions, the density of active sites, ρ , is identical to the survival probability of a spreading simulation, started from a single active site, yielding the relationship^{19,20}:

$$\rho \sim t^{-\alpha} \quad (4)$$

with $\alpha = \delta$. Duality also implies that the critical infection rate λ_c is the same for the two initial conditions. Note that the critical exponents above are not independent, as they are related by the hyperscaling relation $\eta = \frac{d}{z} - \delta - \alpha$, where d is the dimension of the underlying lattice^{19,20}.

Our results indicate that Hebbian (and anti-Hebbian) learning qualitatively changes the emergent phenomena in the CP, leading to a rich class of behavior, depending on the microscopic details of how learning is implemented, as well as on the initial conditions. Our analytic and numerical results show that local incentives can lead to the opposite global effects, like avoiding activated or activating neighbors can help global activity propagation, and vice versa. Negative reinforcement in higher dimensions (like in infection and population spreading) leads to the emergence of two distinct phase transitions, one where a propagating front emerges and another where bulk activity is maintained. Positive reinforcement (motivated by the brain) prevents the formation of an active phase (like quarantining). In higher dimensions, it can lead to a non-universal power-law regime, known as a Griffiths phase, providing an alternative mechanism for this phenomenon. Altogether, our results represent a comprehensive picture of the impact of Hebbian learning on activity spreading, mapping out all universality classes. We expect our main findings to be generic, applicable to a broad range of out-of-equilibrium spreading processes with Hebbian learning in two and higher dimensions. Although our models explore a—to the best of our knowledge—so far overlooked learning mechanism in the CP, the observed family of phenomena fit into the broader class of variants of the CP^{19,21,30}. Some aspects of the collective behavior of our systems are also seen in models with quenched heterogeneities^{31,32}, or adaptation through rewiring^{33,34}. The latter can lead to a rich set of critical phenomena due to the presence of multiple absorbing configurations. We demonstrate that this is also the case for our CP model variants, and, as a frequent feature of such systems¹⁹, the scaling of certain quantities can still be governed by DP, while others deviate from it.

Methods

The CP is controlled by an activation rate λ and a rate μ of spontaneous deactivation, which is usually chosen to be one for convenience. In the numerical implementation, we follow the standard simulation prescription by Dickman³⁵. An active site i is chosen at random from a list of N active sites. The site transitions to the inactive state with probability $1/(1 + \lambda)$ and attempts to activate a neighboring site with probability $\lambda/(1 + \lambda)$. In order to approximate a continuous-time Markov chain, each of these steps increases time by $\Delta t = 1/N$, so that, on average, each active site acts once per unit of time. In the standard CP, if activation is chosen, a neighboring site is chosen at random with uniform probabilities. If this site is inactive, the activation attempt is deemed successful and the neighbor transitions to the active state. For actual biological, social and physical networks, relationships between sites need not be uniform. To model this, we introduce edge weights, carried in a weight matrix, W , where w_{ij} is the weight of the connection from site i to site j . Then, the direction of activation is chosen with probability proportional to the edge weight in that direction. As a simple example, if site i in a 1D chain has chosen to activate a neighbor, it will attempt to activate site $i + 1$ with probability $w_{i,i+1}/(w_{i,i-1} + w_{i,i+1})$ and it will attempt to activate site $i - 1$ with probability $w_{i,i-1}/(w_{i,i-1} + w_{i,i+1})$.

To model Hebbian learning in the CP, we choose a multiplicative type of event-based reinforcement rule: the weights are multiplied with a constant factor r after each successful activation event between sites i and j . This can be realized according to one of the three possible scenarios: mutual, source, or target learning, as illustrated in Fig. 1 and summarized in Table 1. In the mutual learning model, both w_{ij} and w_{ji} are multiplied by the reinforcement parameter, r . In the source learning model, only the weight w_{ij} is multiplied by r , while in the target learning model only w_{ji} changes. In all cases, if site i attempts to activate the already-active site j , w_{ij} and w_{ji} are left unchanged. In line with the usual intuitive picture of Hebbian learning, reinforcement happens in these models only at events when two sites become simultaneously active. However, simultaneous activity alone does not lead to reinforcement, as a direct activation event is required.

Note that even with the application of a reinforcement rule, the total rate of activations sourcing from a given site is invariant (λ), as changes in the weights only affect the preferences according to which a site spends its contact time. In other words, we have an implicit normalization for the total activation rate, and learning induces directional selectivity towards ($r > 1$) or against ($r < 1$) some partners. Without some sort of normalization, Hebbian learning is in general unstable as either all activation rates grow until a maximum allowed value or all rates decay to zero³⁶.

Our implementation of local learning is in stark contrast to past attempts in the literature to capture adaptation in spreading processes. In the traditional framework of adaptive networks^{33,37}, the topology of the connections is changing, while in the more recent approach of “awareness”^{38,39}, the infection rate changes according to the local or global

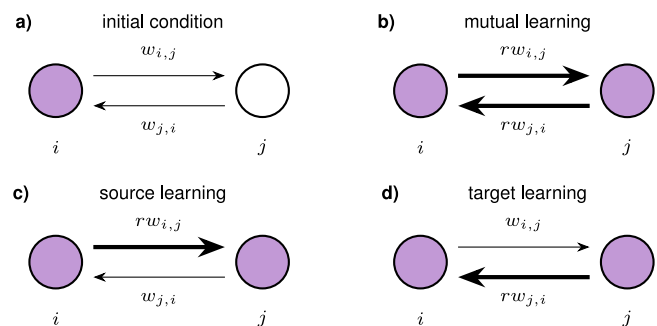


Fig. 1 | Illustration of our Hebbian learning rules in the contact process. (a) denotes the initial condition while (b–d) denote successful activation for the mutual learning, source learning, and target learning models, respectively. Purple denotes an active site and bold denotes a modified connection, multiplied by the reinforcement parameter, r . The edge from node i to node j has initial weight w_{ij} , which is used to calculate activation probability.

Table 1 | Summary of the studied learning models

	1D		2D	
	$r < 1$	$r > 1$	$r \rightarrow 0$	$r \rightarrow \infty$
source	MAC	directional lock	full: MAC, spreading: DyP	GP (exp. localization)
mutual	full: MAC, spreading: ballistic	localization	full: MAC, spreading: DyP	localization
target	ballistic front (*)	localization	full: MAC, spreading: DyP (*)	localization

Gray background color indicates the lack of a phase transition. For $r < 1$, the behavior can depend on the initial condition being a fully active lattice (full) or a single active site (spreading). We use the following abbreviations: DyP: dynamical percolation, GP: Griffiths phase. MAC stands for a broad class of behavior that is driven by multiple absorbing configurations, where certain aspects are governed by the DP universality class, see text. The precise value of the reinforcement parameter r is often irrelevant, apart from whether it corresponds to negative ($r < 1$) or positive ($r > 1$) reinforcement. (*) stands for emergent quenched directional configuration(s) in the $r \rightarrow 0$ limit, see text.

activity. Hence, our implementation of Hebbian learning is a qualitatively different adaptation mechanism that might in practice appear simultaneously with other modes of adaptation, such as rewiring the topology or adjusting the local infection rates based on awareness.

To investigate the effects of our Hebbian learning rules on the CP, we consider two initial conditions—spreading from a single active site or decaying from a fully-active lattice. For spreading simulations, we monitor the number of active sites N , the survival probability P , and the mean square radius R ; while for a fully active initial condition we monitor the density of active sites ρ . Initially, $w_{ij} = 1$ for all neighbors (i, j) . Each edge weight can then be written as $w_{ij} = r^{u_{ij}}$, where u_{ij} is the number of times the connection w_{ij} has been reinforced. In the 1D chain case, the quotient $w_{i,i-1}/(w_{i,i-1} + w_{i,i+1})$ can be written as $1/(1 + r^{n_i})$ where $n_i = u_{i,i+1} - u_{i,i-1}$ is the difference in usage between outgoing edges. Likewise, $w_{i,i+1}/(w_{i,i-1} + w_{i,i+1})$ can be written as $1/(1 + r^{-n_i})$. Hence, in the 1D chain, all relevant information concerning the actions of site i can be stored in the quantity n_i , which we refer to as the imbalance of the site. With this definition, the imbalance of a site is zero if it has activated both of its neighbors the same number of times, indicating no bias in the preferences for future infection events.

We note that the contact process endowed with the above event-based learning rules is still a Markov process, however, on an extended state space: besides the usual occupation variables of the standard CP, the configurations of the system are also characterized by the additional set $\{u_{ij}\}$ (or $\{n_i\}$ in 1D) which encode the actual weights. This model differs from the standard CP in that it has an infinite number of absorbing configurations differing only in the weight variables. As a consequence, our model is exempted from the scope of the DP conjecture mentioned in the Introduction.

Note that to check the emergent behavior for anti-Hebbian values of $r < 1$, it is also useful to investigate the limiting case $r \rightarrow 0$. When tracking the number of times each edge has been used, as r approaches 0, the site will choose to spread activity along the least used edge more often. In the $r \rightarrow 0$ limit, activity always spreads along the least-used edges. If there are multiple such edges, the direction of activity spreading is chosen uniformly randomly, like in the standard CP. Similarly, for Hebbian values of $r > 1$ it is useful to consider the case with $r \rightarrow \infty$, where activity spreads along the most-used edges.

For investigating spreading from a single active site, we chose our lattice size such that the spread never touches the boundary. For example, to run simulations to $t = 10^6$ for mutual learning in 1D, it was sufficient to have lattice sizes of $L = 10^6$ for $r \rightarrow 0$. For the same model in 2D, $L = 2.5 \times 10^4$ was

sufficient up to time $t = 10^5$. We typically collected over 10^4 samples for spreading simulations and around 10^2 for fully infected simulations. In 1D, fully infected simulations had a typical linear size of $L = 10^6$, and in 2D $L = 5 \times 10^3$, but at least $L = 2 \times 10^3$ to reduce finite-size effects.

We regarded the activation rate λ as a control parameter and performed simulations for a series of λ values while the other model parameters were fixed, measuring $P(t)$, $N(t)$, $R(t)$, and $\rho(t)$ as a function of time. Assuming power-law characteristics at criticality, we estimated the critical activation rate λ_c by finding the curve which is closest to a straight line in a log-log plot at late times. The critical exponent was then estimated by a linear fit to this curve in the late-time domain. The error margins were estimated by performing fits to curves with λ closest to λ_c (both below and above λ_c).

Results

Negative Reinforcement in one dimension

In one dimension, anti-Hebbian negative reinforcement ($r < 1$) results in qualitatively different behavior in the three different learning modes as illustrated in Fig. 2. Here, we show the density profiles at various times starting from a single active site at the center. In the source learning model, the activity peaks at the center and its spreading at the critical point is partially compatible with the DP universality class, as can be seen in Table 2. We find that the scaling function of density profiles—plotted in the inset of Fig. 2a—has the same non-Gaussian tail, $\tilde{\rho}(x) \sim e^{-C|x|^\Phi}$ with $\Phi \approx 2.4$ as observed for the standard CP²¹. The critical mutual learning model has a qualitatively different behavior: the activity peaks at two ballistically advancing front regions as shown in Fig. 2b. In stark contrast, there is no active bulk phase in the target learning model, Fig. 2c, with only two narrow fronts propagating outwards ballistically that have a finite lifetime.

In the following, we first present detailed scaling results for the critical source and mutual learning models, supported by analytic arguments and further measurements in the Supplementary Note 1 on the behavior of the imbalance variable. Then, we also shed light on the absence of active phase in the target learning model. The case of positive reinforcement ($r > 1$), when there is no active stationary state irrespective of the value of λ , is discussed for all one-dimensional variants of the model in Supplementary Note 2.

Source learning model. In the source learning model, we have explored several values of the learning rate r , leading to qualitatively similar results for all $r < 1$ values. Here, as an illustration, we focus on presenting the results at $r = 0.8$, shown in Fig. 3, with additional results summarized in Table 2. We find that the critical point of the standard CP is inactive with negative reinforcement, and the critical point is monotonically shifted towards larger values of λ as r decreases. Note that duality is violated in this model, as evident from the different values of α and δ , although the transition happens at the same λ_c for both initial conditions. Yet, in terms of standard observables (Fig. 3), all $r < 1$ cases are similar in the sense that within the error they cannot be distinguished from the DP universality in terms of α , z , and $\eta + \delta$. Following directly from the definitions, $\eta + \delta$ governs the total number of active sites in the surviving runs as $N_s \sim t^{\eta+\delta}$. This means that the observables conditioned on survival, N_s and R show DP critical behavior, but the survival probability itself is affected by reinforcement; see the altered values of δ in Table 2. This entails the change of η , characterizing the number of active sites in all trials. The latter two exponents seem to show hardly any change with the reinforcement parameter.

It is useful to interpret these results in the broader context of other CP variants in the literature. For example, the surface critical behavior is also known to be characterized by a modified order parameter exponent, while the exponents related to observables conditioned on survival are unaffected²¹. Further examples to this type of critical behavior are various one-dimensional processes such as the model in ref. 30, which have in common that they possess an infinite number of absorbing configurations, just like our model. For this reason, we refer to this diverse class of critical behavior with multiple absorbing configurations as “MAC”¹⁹. At a MAC,

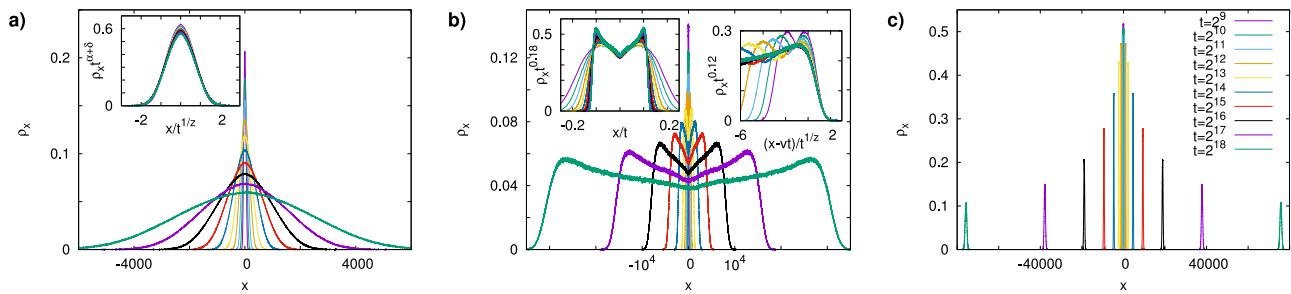


Fig. 2 | Density profiles of the three different learning models in one dimension. Here, we illustrate the ρ_x activity density profiles at various times starting from a single active site in the center ($x = 0$) for the (a) source ($r = 0.8$), (b) mutual ($r = 0.5$), and (c) target ($r = 0.8$) learning models. The source and mutual learning models are at

their critical points, see Table 2, while for the target learning $\lambda = 5$. The insets show a good data collapse of the profiles with the critical exponents listed in Table 2. For the mutual learning model (b), the velocity of the front is $v = 0.11$ and the DP value of z is used in the inset.

Table 2 | Critical scaling in one dimension

	r	λ_c	α	δ	η	z
DP^{51,62}	1.0	3.2979	0.1595	0.1595	0.3137	1.5808
source	0.9	3.512(1)	0.15(1)	0.05(1)	0.45(1)	1.54(3)
	0.8	3.721(1)	0.14(1)	0.04(1)	0.46(2)	1.54(2)
	0.5	4.425(2)	0.15(1)	0.03(2)	0.48(2)	1.53(4)
	0	8.13(1)	0.16(1)	0.04(2)	0.45(2)	1.55(3)
mutual	0.9	3.269(1)	0.29(7)	0.00(1)	0.73(4)	1.005(3)
	0.8	3.248(1)	0.30(8)	0.00(1)	0.75(3)	1.002(4)
	0.5	3.203(1)	0.25(7)	0.00(1)	0.79(4)	0.998(7)
	0	3.193(1)	0.17(5)	0.00(1)	0.83(3)	1.00(2)

Critical activation rate and critical exponents at various negative reinforcement rates ($r < 1$), corresponding to anti-Hebbian learning. For comparison, results on standard directed percolation (DP) are shown up to four digits after the decimal.

certain exponents and exponent combinations are governed by DP, while others (here δ and η) deviate from DP, similarly to ref. 30.

A deeper understanding of our manifestation of the MAC behavior can be obtained through the imbalance variable. As introduced earlier, the imbalance of a site $n_i(t)$ is a signed sum of successful activation events from site i , in which each activation to the left (right) corresponds to a term -1 (1). As shown in Supplementary Fig. 1b, the imbalance distributions tend to a limit distribution at late times, which has Gaussian tails $p_n \sim e^{-Cr^2}$ with an r -dependent constant C for $|n| \gg 1$. This is found to hold irrespective of whether the model is in the active phase or at its critical point. The emergence of a Gaussian tail can be explained by a mean-field treatment, which is presented in detail in Supplementary Note 1. In this calculation, the local states, which are specified by the occupation and the imbalance variables, are assumed to be uncorrelated at different sites. This allows us to write down a master equation for the dynamics of local states, which turns out to reduce to that of a one-dimensional random walk in the presence of a site-dependent potential U_n . Here, the position n of the walker corresponds to the imbalance variable, while the potential is found to be quadratic in n for large $|n|$, as $U_n \simeq n^2 \frac{\ln(1/r)}{2}$. Upon increasing r , the potential changes its character at $r = 1$ from attractive to repellant near the origin. As a consequence, the distribution of imbalance variables will be stationary at late times only if $r < 1$, in which case it has a Gaussian tail: $p_n \sim e^{-U_n} \simeq e^{-n^2 \frac{\ln(1/r)}{2}}$. We note that, although the functional form is correctly reproduced by the mean-field approach, the numerical value of the constant $\ln(1/r)/2$ deviates from the estimate obtained by simulations; for details, see the “Stationary state” subsection of Supplementary Note 1: Source learning model.

The Gaussian tail of the limit distribution of imbalance variables translates to a log-normal tail of the distribution of instantaneous weights. It is worth mentioning that the local variable n_i at a fixed site has the same limit distribution as the ensemble of local variables, and this limit distribution is symmetric under $n \rightarrow -n$.

Mutual learning model. Just like in the source learning case, we obtained qualitatively similar results for all values of $r < 1$, illustrated here through the case of $r = 0.9$, see Fig. 3, with all results summarized in Table 2. As opposed to source learning, the critical activation rate weakly decreases with decreasing r . Thus the $r < 1$ mutual learning model is active at the critical point of the standard CP, indicating that mutual negative reinforcement promotes activity spreading. The other qualitative difference compared to source learning is that the survival probability $P(t)$ tends to a positive limit at late times even at the critical point. This means that regarding, as usual, the limiting value of $P(t)$ as an order parameter, the transition is discontinuous, with a formally vanishing exponent, $\delta = 0$. The other measured critical exponents also differ from those of the source learning model. The dynamical exponent is close to 1 for all $r < 1$, indicating a ballistic spreading of the active zone, in agreement with the density profiles in Fig. 2b. The asymptotic velocity of spreading is found to increase with decreasing r . The number of active sites N shows a power-law like scaling with a much larger η exponent than for the standard CP or the source learning model. The global density $\rho(t)$, when the process starts from a fully active initial state, decays algebraically in time, and the exponent α is close to the DP value for not too high values of r , otherwise it has a somewhat increased value compared to DP. Duality of the DP class is clearly broken—even the corresponding critical exponents are different since $\alpha \neq \delta$. Mutual learning in one dimension thus serves as another example of MAC, a mixture of DP-like and non-DP behavior, although it is clearly different from the MAC manifestation of the source learning model.

The inspection of the density profiles shown in Fig. 2b, as well as the measured values of dynamical critical exponents lead us to the following picture of activity spreading at the critical rate with mutual learning. At a given time, the system can be divided in three parts. An intact region, which has not yet been visited by activity, a front region with a high $[O(1)]$ local density of activity, and an interior part in which sites were reactivated many times so that the imbalance distribution is close to its limiting distribution. It is easy to see that the mutual learning rule with $r < 1$ facilitates activation processes from the actual front particle toward the intact region. When starting from a single active site, as activity spreads outwards, the edge connecting the newly active site back towards the center of the cluster is weakened. When it is this newly activated site's turn to activate others, it is significantly more likely to activate outwards, towards inactive sites. This reduces the amount of times activation is attempted but failed, promoting spreading.

This mechanism results in a ballistic drift of the fronts, at which the process is locally supercritical. In the interior part, the process is critical and the local density decays algebraically in time. This ballistic expansion of the critical zone (c.f. the sublinear creeping of the front $R(t) \sim t^{1/2}$ in the standard CP) ensures that the process remains surviving with a finite probability even at the critical point. Specifically, at a fixed site x , the local density is $O(1)$ at time $t_x \simeq x/v$, i.e., the time at which the front sweeps through site x . Here, v denotes the mean asymptotic velocity of the front. After this event, the local density decreases in time asymptotically as $\rho_x(t) \sim (t - x/v)^{-\alpha}$. Summing up these local densities in the interior part $[-vt, vt]$, we obtain an algebraic

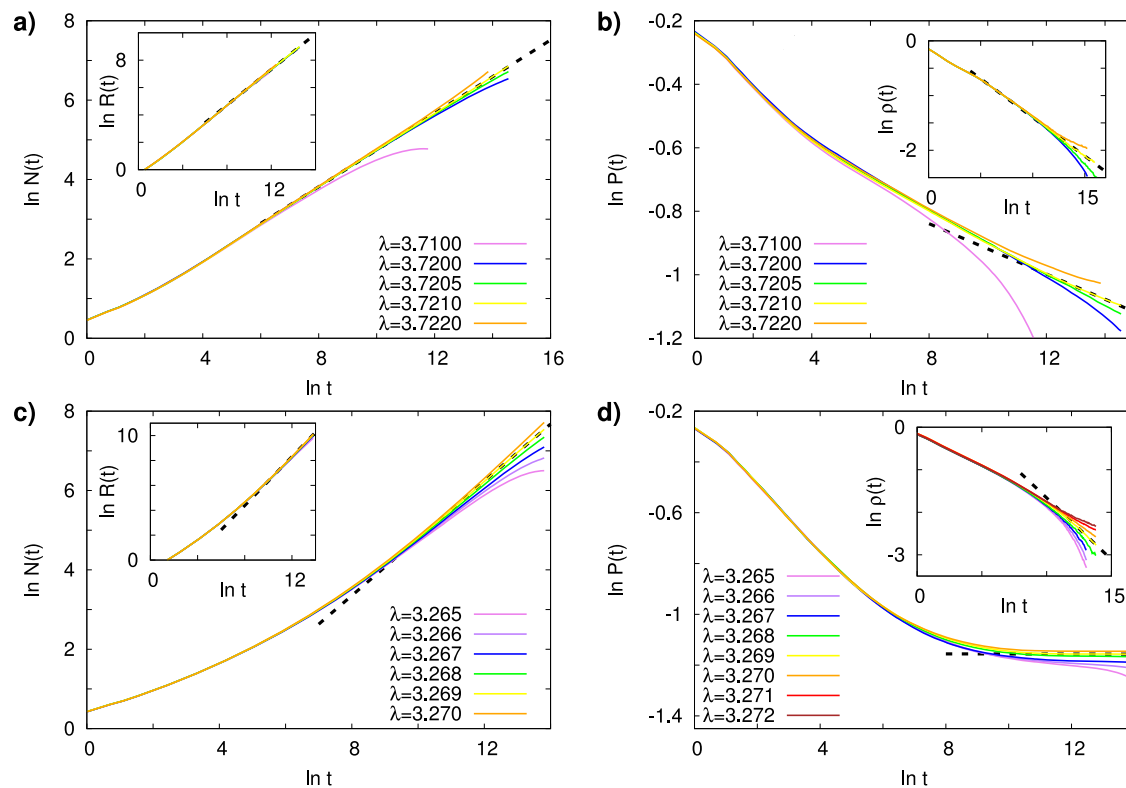


Fig. 3 | Critical scaling with negative reinforcement in one dimension. Source learning is shown in the top panels (a, b), with mutual learning indicated in the bottom panels (c, d). In the main panels, for varying activation rates λ , we illustrate the number of active sites N (left panels: a, c), and the survival probability P (right

panels: b, d). The insets show the mean square radius R (left panels: a, c), and the density of active sites ρ (right panels: b, d) at $r = 0.8$ (source learning) and $r = 0.9$ (mutual learning). The asymptotic critical behavior is illustrated by the straight dashed line.

increase of the number of active sites $N(t) \sim t^{1-\alpha}$. Assuming that the bulk criticality belongs to the DP universality class, we obtain $1 - \alpha = 0.8405$. Numerical estimates presented in Table 2 are somewhat lower; this deviation may be attributed to corrections stemming from the front region which is found to spread out proportionally to $t^{1/2}$ with the DP dynamical exponent z on top of the ballistic advancement, see Fig. 2b.

As shown in Supplementary Fig. 2, the distribution of the imbalance variables tends to a stationary distribution with exponential tails, $p_n \sim e^{-a|n|}$, with $a = 0.87(2)$. This scaling form, apart from the numerical value of a , is in agreement with our mean-field calculation, shown in the “Stationary state” subsection of Supplementary Note 1: Mutual learning model. Here, just as for the source learning model, the dynamics of imbalance variable is represented by a one-dimensional random walk in a site-dependent potential U_n . Again, we find this random walk to be recurrent, having a stationary state only if $r < 1$. In this case, the potential has a linear asymptotics $U_n \simeq |n| \ln 3$ for large $|n|$, leading to exponential tails of the distribution of imbalance variable as $p_n \sim e^{-U_n} \sim 3^{-|n|}$. Note that the exponential tail of imbalance variable corresponds to a power-law tail of the instantaneous weights.

Target learning model. In the target learning model with negative reinforcement ($r < 1$), simulations started from the fully active state show that the density decays to zero at late times, no matter how large λ is. This means that in this case, there exists no active stationary phase. The reason for this lies in the dynamical rules of this model which enforce a “persistence” of the directionality of activity spreading. This means that, after a successful activation event in some direction, the target site will acquire a bias in the activation rates to the same direction. As a consequence, during the time evolution of the process, domains are spontaneously formed within which sites have the same directionality i.e., imbalance variables are all positive or all negative, and this bias of domains is steadily

amplified in the presence of activity. Naturally, the activity spreading in a biased environment ($n > 0$ or $n < 0$) is characterized by a non-zero drift velocity. In case of a total bias ($n \rightarrow \infty$ or $n \rightarrow -\infty$), also known as the one-sided CP⁴⁰, the survival of a steadily drifting bunch of activity is possible only in an infinite system. Therefore, an environment composed of alternating biased domains can not sustain activity for arbitrarily long time. This conclusion is compatible with the result of mean-field approach in Supplementary Note 1: Target learning model, stating that there is no stationary state of the imbalance variable for any value of r .

In the case of starting the process from a single active seed, the activity spreads out ballistically in two directions leaving behind a predominantly left and right-oriented environment on the two sides of the origin. However, as opposed to the fully active initial condition, the fate of spreading activity will be different for $0 < r < 1$ and for $r \rightarrow 0$. In the former case, there is a small but finite probability that in the front region, where the imbalances are close to zero, an oppositely biased domain emerges by a rare fluctuation. This domain hinders the drift of activity behind it and may ultimately lead to the vanishing of activity. Our numerical simulations (not shown) indicated that such catastrophic fluctuations occur for any $0 < r < 1$ and for high values of λ , suggesting the absence of an active phase for $\lambda > 0$.

In the limit $r \rightarrow 0$, however, the probability for the appearance of an oppositely biased site tends to zero, and the process reduces to the one-sided CP. This model has an absorbing phase transition at $\lambda_c = 3.306(2)$, which belongs to the DP universality class when inspecting it from a co-moving frame⁴⁰.

Higher dimensional systems

In higher dimensions, for simplicity, we restrict our presentation to the results obtained in the limits $r \rightarrow 0$ and $r \rightarrow \infty$. These limits are computationally more feasible to study, lead to less severe scaling corrections, and are usually expected to be representative of the generic cases with negative

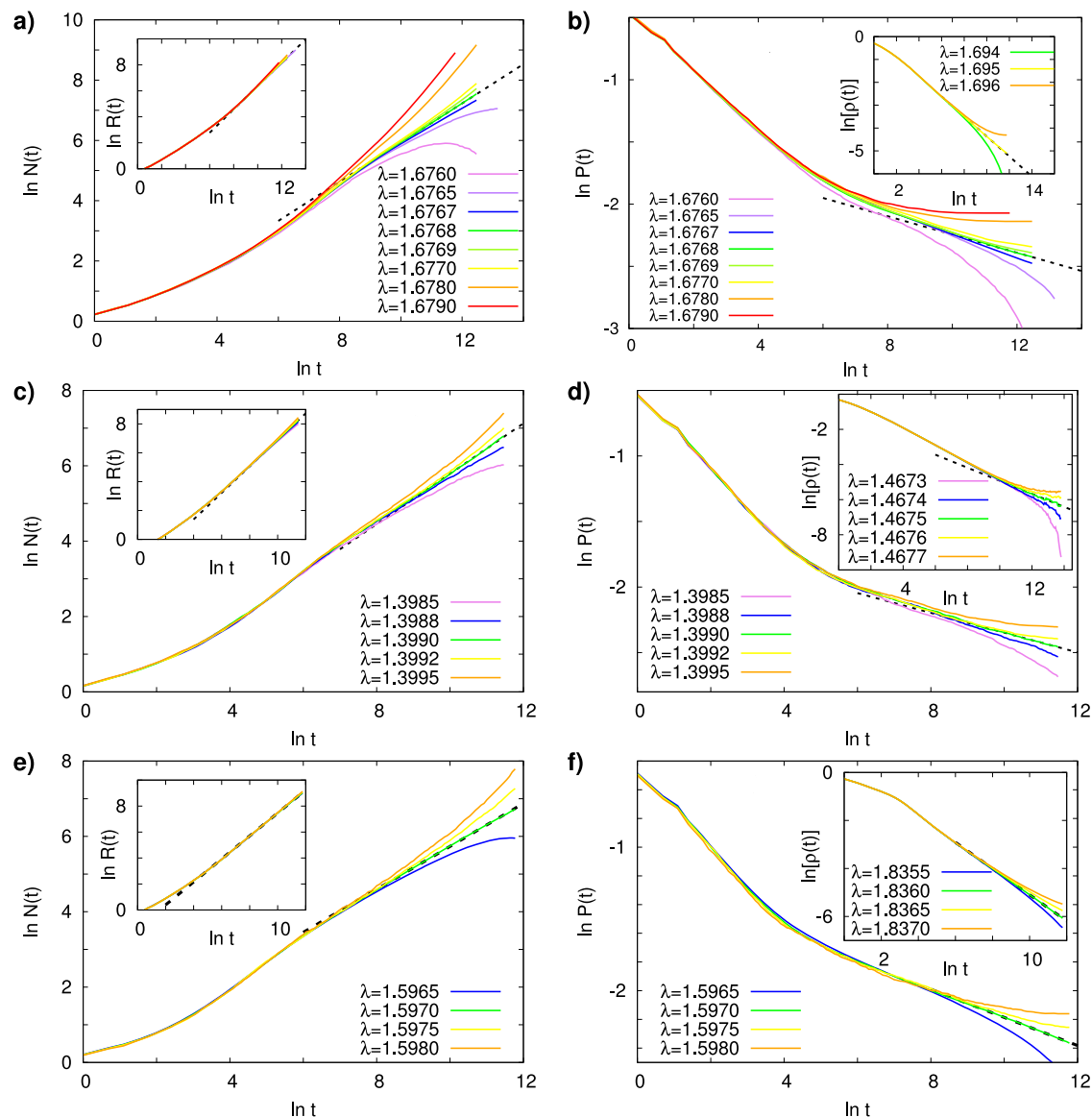


Fig. 4 | Hebbian learning in 2D. Temporal scaling behavior for varying activation rates λ is shown for source learning (a, b), mutual learning (c, d), and target learning (e, f). In the main panels we illustrate the number of infected sites N (left panels: a, c, e), and the survival probability P (right panels: b, d, f). The insets show the mean

square radius R (left panels: a, c, e), and the density ρ for $r \rightarrow 0$ (right panels: b, d, f). The asymptotic critical behavior is illustrated by dashed lines either at the spreading critical point, λ_1 , for N , P , and R ; or at the bulk critical point, λ_2 , for ρ , see Table 3.

($r < 1$) and positive ($r > 1$) reinforcement, respectively. As in 1D, the target learning model is an exception at $r \rightarrow 0$, as in this limit the heterogeneities become frozen, leading to an emergent quenched disorder, as discussed later in this section.

As opposed to the one-dimensional models, a generic observation for the two-dimensional variants of these models is that they possess distinct phase transition points for different initial conditions. In the case of spreading from a single active seed, the transition point can be identified by inspecting $P(t)$, $N(t)$, and $R(t)$. The critical activation rate in this case is denoted by λ_1 . For a fully active initial state, the transition point can be found by inspecting the global density $\rho(t)$, and the corresponding bulk critical rate, which is higher than λ_1 , is denoted by λ_2 .

Negative reinforcement. For all cases, our simulation results are illustrated in Fig. 4 for $r \rightarrow 0$, with the estimates of the critical exponents listed in Table 3. In the source learning model, for a fully-active initial condition, $\lambda_2 \approx 1.695$ and α is around the DP value, while spreading simulations indicate a lower critical rate around $\lambda_1 \approx 1.6768$. In the

mutual learning model, we again observe two critical points, with the exponents presented in Table 3. Our results indicate that the system is moved to a supercritical state when run at the clean critical point but with $r < 1$. Both the spreading critical point (λ_1) and the bulk critical point (λ_2) are shifted down for $r < 1$ negative reinforcement, just like in the 1D case.

A similar scenario of distinct phase transitions is observed for the target learning model. Here, λ_1 is below while λ_2 is above the clean critical point. For this model with $r \rightarrow 0$, a special spontaneous “freezing” of random directions takes place, which is analogous to the extraordinary behavior of the same model in one dimension. Here, the set of least-used (highest-weight) directions at each site constitutes the “directional configuration” of the system. Occupation variables then follow a simple dynamics that is completely determined by the directional configuration of the system. It is easy to see that there is an infinite number of directional configurations that remain invariant under the dynamics of the model. The only restriction for creating such invariant configurations is non-reciprocity, i.e., whenever site i has a least-used direction to site j then site j must not have a least-used direction to site i . A reciprocal situation is unstable against the dynamics and

Table 3 | Critical scaling in 2D

	λ_1 (spread)	λ_2 (bulk)	α	δ	η	z
DP ⁵²	1.64874(2)	1.64874(2)	0.4526(7)	0.4526(7)	0.233(6)	1.757(8)
DyP ^{63,64}	4.66571(3)	4.66571(3)	0.092	0.092	0.586	1.1293
source	1.6768(1)	1.695(1)	0.44(1)	0.07(7)	0.65(3)	1.11(2)
mutual	1.3990(2)	1.4675(1)	0.44(4)	0.07(4)	0.71(7)	1.08(3)
target	1.5970(5)	1.8360(5)	0.55(4)	0.099(4)	0.57(4)	1.13(2)

Critical activation rates and critical exponents at negative reinforcement rate $r \rightarrow 0$, in comparison to the directed percolation (DP) and dynamical percolation (DyP) universality classes. The exponents δ , η , and z pertain to the spreading critical point at λ_1 , while α pertains to the bulk critical point at λ_2 .

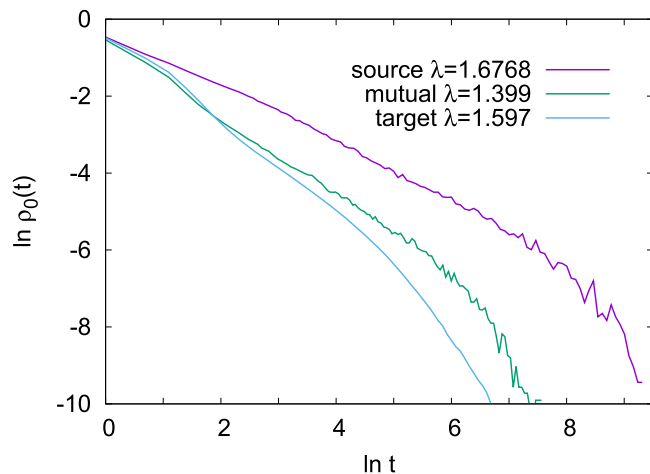


Fig. 5 | Local activity scaling in 2D. Time dependence of the local density ρ_0 at the origin ($x = 0$) in the $r \rightarrow 0$ limit at critical spreading activation rate λ_1 (see Table 3) for source, mutual, and target learning. In all cases, we observe a sharp decay at long enough times, indicating a subcritical bulk phase.

can only happen if the local activity ceases to exist before fixing the direction of the link. Indicating the least-used directions by arrows on links, invariant configurations are constructed by placing arrows (in either direction) on links or not placing any, under the condition that each site has at least one outgoing arrow. According to our numerical observations, the $r \rightarrow 0$ target learning process—when started from an isotropic directional configuration—evolves under the dynamics to an invariant directional configuration, which is chosen randomly by the fluctuating dynamics. Obviously, the evolution of directional configuration may also freeze locally in some domain before settling into an invariant configuration by the vanishing of activity in that domain. This quenching phenomenon is similar to what happens in one dimension, although the directional configuration in that case is trivial due to the 1D topology.

In all cases, our observations imply the existence of two phase transitions in these models, dividing the emergent behavior into three distinct phases of (i) no activity below λ_1 , (ii) a propagating front with no bulk activity for $\lambda_1 < \lambda < \lambda_2$, and (iii) an active bulk phase for $\lambda > \lambda_2$. As a further evidence for this picture, at λ_1 we eventually observe a fast decay of the local density at the origin as illustrated in Fig. 5, indicating that the bulk is asymptotically inactive at λ_1 . The λ_2 location of the bulk critical point is different in all three models, as identified by the critical point of fully infected simulations based on the scaling of the total density $\rho(t)$.

The source and mutual learning models are close to the DP scaling at the bulk critical point, λ_2 . In the case of the target learning model, for which spontaneous freezing of directions occurs (c.f. the fluctuating weights in the other two models), the exponent α differs significantly from the DP value, although strong corrections may not be excluded. At the spreading critical point λ_1 , we see a marked deviation from the DP expectations, close to the values of the dynamical percolation (DyP) universality class. In this case,

target learning is closest to the DyP exponents, but the other two model variants might also show the same behavior (as suggested by z and δ), albeit with stronger corrections (mostly in the case of η). The DyP is known to capture the phase transition in the susceptible-infected-recovered (SIR) model. Note that the DyP exponents satisfy a modified hyperscaling relation²¹, given by $\eta = d/z - 2\delta - 1$. This hyperscaling relation is also satisfied by our exponents for each of the three learning models at λ_1 . Next, we argue that indeed, at λ_1 , asymptotically all three learning models are expected to follow SIR dynamics.

As in the 1D model with mutual learning, we can distinguish three different domains of the spreading system above λ_1 . These are the (i) ballistically expanding annular front region, where the activity is high, (ii) the unvisited domain outside of the front, and (iii) the interior part which has already been swept through by the front. Spreading of the front region toward the outer empty region where there is no directional bias is facilitated compared to spreading in the inner domain, in which there is a random directional field for transmitting activity. As a consequence, for a not too high values of λ , the process is still locally subcritical in the interior, although the front region is steadily expanding outwards. Note that once activity visited a local region, the homogeneity of weights there is irreversibly broken. This suggests an analogy with the dynamics of the SIR model, which differs from the standard CP in that active sites do not return to the inactive state but get removed irrevocably from the system. The analogy is salient: the unvisited (homogeneous) local regions correspond to the susceptible state of the SIR model, whereas the visited (inhomogeneous) local regions correspond to the recovered state. This correspondence is obviously not perfect: in the SIR model, recovered sites can never be reactivated, while in our model, this rule is less strict, since activity is able to penetrate into an inhomogeneous region. Nevertheless, as this latter region is subcritical, the penetration length is finite. Hence, concerning large-scale properties at the critical point, this circumstance is irrelevant. Based on this observation, we expect the lower (spreading, λ_1) critical points of our two-dimensional learning models to belong to the DyP universality class of the SIR model.

Thus, the lower phase transition of the model (λ_1) is governed by the criticality of the front region, while the interior of the system is subcritical at this point. It is clear that such a situation can be realized only in dimensions higher than 1, in which the boundary between the intact and visited domain has a dimension $d - 1 \geq 1$. On the contrary, in one dimension the criticality of the system cannot be sustained solely by a (zero-dimensional) front. As a consequence, in 1D models, the entire inner region must be critical at the spreading transition. This is in agreement with our numerical observations that our models in one dimension display only a single phase transition i.e., $\lambda_1 = \lambda_2$ in 1D.

Positive reinforcement. As found in Supplementary Note 2, and illustrated in Supplementary Fig. 3, in the one-dimensional case within a mean-field treatment, there is generally no active phase with positive reinforcement. This is also the case in higher dimensions. While the mutual and target learning models lead to a quick localization of the activity for all $r > 1$, the source learning case can show a qualitatively different behavior in higher dimensions. This generic phenomenon is easier to understand for $r \rightarrow \infty$. In this case, the direction of the first

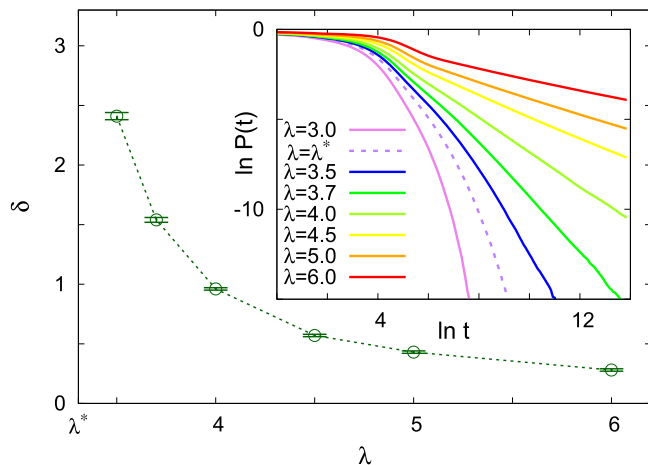


Fig. 6 | Griffiths phase in the 2D source learning model with positive reinforcement. In the inset, we plot survival probability P as a function of time for varying activation rates λ . In this representation, straight lines indicate a Griffiths phase of non-universal power-laws, expected to appear above the critical point of the one-dimensional totally asymmetric CP, $\lambda^* = 3.306(2)$ (dashed line)⁴⁰. Here, the Griffiths phase is a manifestation of the “ant mill” phenomenon, where the activity propagates along randomly formed loops until it ceases to exist⁴¹. The main panel indicates the best fit and the estimated error of the δ exponents that correspond to the non-universal power-laws observed in the inset for $\lambda > \lambda^*$.

successful activation from a given site “freezes” i.e., any further activation from that site can occur only in that direction. This means that learning leads to an emergent, correlated quenched disorder in the system. As a consequence, in spreading simulations, a train of particles is forming, the front particle performing a random walk until it revisits a site that was activated before. In any dimensions, this happens with probability one—after a finite transient—and then the activity can only propagate around the one-dimensional loop. This is a manifestation of the “ant mill” phenomenon⁴¹. An ant mill emerges when certain ants reinforce their trajectories by pheromones following each other until they die of exhaustion when accidentally forming a loop. The simplest mathematical model considers multiplicative reinforcement in the form of a directed edge reinforced random walk, with rigorous results in all $d \geq 2$ dimensions⁴².

Large loops occur with an exponentially small probability, but they are active for an exponential amount of time above the critical point of the one-dimensional one-sided CP with $\lambda^* = 3.306(2)$ ⁴⁰. Hence, above λ^* , we observe a non-universal power-law decay of the survival probability as shown in Fig. 6 in the large r limit. Note that non-universal power-law dynamics is well-known in quenched disordered systems such as the disordered CP in the inactive phase, where disorder implies the presence of so-called rare regions with above-average couplings—a phenomenon known as a Griffiths effect^{32,43–45}. However, unlike in the CP with quenched disorder where rare regions are present *ab ovo*, in our learning model, the rare regions (loops) are formed by the dynamics itself in the early stage of the process. The emergence of such a Griffiths phase is expected to be a generic result that holds for all $r > 1$ reinforcement rates in all $d \geq 2$ dimensions above λ^* . Note that in one dimension, loops have only two sites, so a Griffiths phase with the same mechanism cannot occur.

Discussion

In this study, we introduced generalizations of the contact process with a local learning rule inspired by Hebbian learning, leading to rich emergent phenomena. We found that local incentives can lead to the opposite global effects, like locally avoiding previously activating or activated neighbors can turn the inactive phase into a globally active phase, and vice versa. For example, positive reinforcement in the mutual learning model is akin to quarantining simultaneously exposed groups. In this case, if infection is

passed between two sites, they preferentially interact with each other, instead of their other neighbors, leading to activity localization at a finite scale that depends on the value of the reinforcement parameter, $r > 1$. Such localization hinders activity spreading at the global scale, forcing the system to be in the inactive phase.

In stark contrast to standard critical phenomena, microscopic differences in the learning models can lead to dramatic changes in the emergent behavior, from completely destroying a phase transition to introducing an additional phase transition. For example, in two dimensions, with anti-Hebbian negative reinforcement, we see different phase transitions depending on the initial conditions. At the fully infected (bulk) critical point, we observe manifestations of a class of behaviors controlled by multiple absorbing configurations (MAC) behavior, where some observables follow the DP scaling, while (depending on the model) others show a distinct behavior affected by learning. At the spreading critical point, in higher dimensions, our results suggest a critical point in the universality class of dynamical percolation (DyP), as in the SIR model. In this case, learning with negative reinforcement turns the SIS model (without immunization) into an effective SIR model, even though the local infection and healing rates remain unaffected. In the region between the spreading and the bulk critical point, the system is in a phase showcasing a propagating front with a subcritical bulk. It is an interesting future direction to explore if such a phase could be observed in some real-life applications, either in infection, brain activity or population spreading. Although the contact process is a genuinely non-equilibrium model and learning leads to changes in the model parameters coupled to the dynamics, we were also able to obtain informative analytic results, further supporting our simulation observations.

Our minimal Hebbian models with positive reinforcement have been inspired by reinforcement learning in the brain. In the source learning model, positive reinforcement leads to a Griffiths phase in higher dimensions. There is increasing evidence that brain dynamics also shows hallmarks of being in a Griffiths phase^{46–49}, believed to be advantageous in terms of memory and computational capabilities⁹. Signatures of Griffiths phases might be also present in artificial neural networks⁵⁰, an exciting future direction to explore in relation to Hebbian learning. Note that, in standard models, a Griffiths phase is either caused by time-invariant (quenched) disorder⁴⁴, or a hierarchical-modular architecture of cortical networks^{9,51}. Our results serve as an independent alternative mechanism for a Griffiths phase, a manifestation of the “ant mill” phenomenon⁴², originating from dynamical trapping in activity loops emerging from heterogeneities induced by learning. It remains an open problem—in both real and artificial neural networks—to unveil which sort of mechanism is underlying the observations of Griffiths effects.

The target learning model leads to an emergent quenched system in the $r \rightarrow 0$ limit, leading to invariant directional configuration(s) in both 1D (totally asymmetric CP⁴⁰) and 2D. Note that in 2D the system is disordered, as there are infinitely many directional configurations depending on the initial condition and the dynamics. The directional configuration also contains correlations resulting from the generating dynamics. Without learning, uncorrelated quenched disorder in the rates leads to a distinct type of critical behavior characterized by ultra-slow activated scaling⁴⁴ below four dimensions^{52,53} and on certain types of complex networks⁵⁴, in line with the perturbative argument known as the Harris criterion⁵⁵. It is an interesting open question to explore whether such a directional quenched disorder is a relevant perturbation at the critical point, potentially leading to exotic, strong disorder phenomena.

To the best of our knowledge, our work presented here is the first study of Hebbian (and anti-Hebbian) learning for activity spreading, with several future directions remaining open. Such extensions could consider alternative forms of Hebbian learning, either incorporating information on the duration when the sites are simultaneously active, or different normalization schemes. For example, in the case of positive reinforcement, brain dynamics would suggest the study of the potentially stabilizing impact of inherently decaying activation rates³⁶. As real-life infection spreading is complicated, combinations with other mechanisms, like the simultaneous impact of

Hebbian learning and awareness^{38,39}, or excitatory and inhibitory connections⁵⁶, are especially of interest as these are expected to lead to an even more rich phase diagram.

Although the CP is equivalent to the SIS model on a lattice, this is no longer true on complex networks. In stark contrast to the CP⁵⁷, in the SIS model there is no inactive phase on scale-free networks with degree exponent $\gamma > 2$ ^{58–60}. Hence, positive reinforcement might not completely eliminate the active phase. Instead, a phase transition might happen between an inactive and an active phase on such networks. Another direction of study is unveiling the condition under which a similar reinforcement-driven Griffiths phase occurs on complex networks through the ant mill phenomenon⁴².

Data availability

The data for the plots can be accessed at Zenodo <https://doi.org/10.5281/zenodo.19237533>.

Code availability

Code used to simulate the models in this study is available from the corresponding author.

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Author contributions

W.T.E. and R.J. contributed equally. I.A.K. designed and supervised the research; R.J. developed the mean-field description; W.T.E. and R.J. performed the numerical simulations; all authors analyzed data and wrote the paper.

Competing interests

The authors declare no competing interests.

Additional information

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